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**DEPARTMENT OF INTERNET OF THINGS AND ROBOTICS ENGINEERING (IRE)**

FACULTY OF CYBER PHYSICAL SYSTEM ENGINEERING

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# LAB REPORT 01

## DIGITAL WALLET BACKEND SYSTEM

*(Object-Oriented Programming with Encapsulation & Inheritance in Dart)*

**COURSE CODE: IRE 460**

**COURSE TITLE: MOBILE PLATFORM FOR IOT DEVICES SESSIONAL**

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# 1. Problem Description

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This lab task involves creating a simple backend system for a Digital Wallet application in Dart. The application manages two forms of bank accounts: **Standard Accounts** and **Premium Accounts**. The core business logic specifies that premium account holders receive an automatic cash-back bonus whenever funds are deposited into their account.

# 2. Architecture and Technical Requirements

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1. **Encapsulation (`BankAccount`):**
   - Create a base class named `BankAccount`.
   - Store balance in a private variable (`_accBalance`).
   - Provide a public getter (`get balance`) to read balance without direct external modification.
   - Implement a `deposit(double amount)` method to handle fund deposits safely.
2. **Inheritance and Polymorphism (`PremiumAccount`):**
   - Implement a derived class `PremiumAccount` that extends `BankAccount`.
   - Override `deposit(double amount)` so that premium customers automatically receive an additional **5% cash-back bonus** on every deposit.

# 3. Design and Logic Explanation

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- **Encapsulation Mechanism:** By applying the underscore prefix (`_accBalance`), the field is scoped library-private. External code cannot directly overwrite the balance, ensuring validation rules remain intact.
- **Inheritance Model:** `PremiumAccount` reuses the base balance management logic from `BankAccount` without redundant code.
- **Polymorphic Bonus Addition:** In `PremiumAccount`, the `deposit` method calculates the cash-back bonus:

  $$\text{Cashback} = \text{amount} \times 0.05$$

  The total amount credited to the account becomes amount + Cashback, which is handed over to `super.deposit(amount + cashback);`.

## 4. Source Code Implementation (digital_wallet.dart)

```dart
// Base Account Class
class BankAccount {
  double _accBalance = 0.0; // Private balance variable

  // Public getter for reading balance
  double get balance => _accBalance;

  void deposit(double amount) {
    if (amount <= 0) {
      print("Invalid deposit! Amount should be positive.");
      return;
    }
    _accBalance += amount;
    print("Deposited: \$$amount | Available Balance: \$$_accBalance");
  }
}

// Premium Account Class inheriting BankAccount
class PremiumAccount extends BankAccount {
  @override
  void deposit(double amount) {
    if (amount <= 0) {
      print("Invalid deposit! Amount should be positive.");
      return;
    }
    // Calculate 5% cashback
    double cashback = amount * 0.05;
    print("[Cashback] 5% bonus added: \$$cashback");
    super.deposit(amount + cashback);
  }
}

void main() {
  print("=== Testing Standard Account ===");
  var myAccount = BankAccount();
  myAccount.deposit(2000.0);
  print("Current balance: \$${myAccount.balance}\n");

  print("=== Testing Premium Account ===");
  var myPremium = PremiumAccount();
  myPremium.deposit(2000.0); // 2000 + 100 cashback = 2100
  print("Current balance: \$${myPremium.balance}");
}
```

## 5. Execution Output and Results

Executing the Dart script produces the following terminal output:

```
=== Testing Standard Account ===
Deposited: $2000.0 | Available Balance: $2000.0
Current balance: $2000.0

=== Testing Premium Account ===
[Cashback] 5% bonus added: $100.0
Deposited: $2100.0 | Available Balance: $2100.0
Current balance: $2100.0
```

## 6. Discussion and Conclusion

- **Standard Account Behavior:** A deposit of $2000.0 adds exactly $2000.0 to the balance, accessible only via the getter property.
- **Premium Account Behavior:** A deposit of $2000.0 triggers the overridden method, which adds a 5% cash-back bonus of $100.0 ($2000 \times 0.05$). The total credited amount is $2100.0.
- **Conclusion:** The system accurately implements the core OOP concepts of encapsulation, inheritance, and polymorphism in Dart with modular, error-checked code.